

OPERATION FALSE ASSURANCE

Team Name	BYEBUST
Phase Number / Title	Phase 4: Interactive Control and Long-Lived External Sessions (P4-04)
Incident Date	2025-10-08
Date Submitted	2026-08-22

No.	Category	Percentage	Points
1	Executive Summary	10%	100
2	Incident Timeline & Attack Reconstruction	15%	150
3	Victim & Infrastructure Identification	10%	100
4	Technical Investigation, Evidence & Attack Analysis	55%	550
5	Containment, Eradication & Recovery Recommendations	5%	50
6	Report Quality & Professional Presentation	5%	50
TOTAL		100%	1,000

1. EXECUTIVE SUMMARY

During Phase 4 of Operation False Assurance (Interactive Control and Long-Lived External Sessions), threat hunting was conducted across network metadata to analyze outbound command-and-control (C2) channels established by the compromised host `10.11.28.101`. Analysis of Zeek SSL/TLS handshake metadata (`zeek.ssl`) and DNS resolution streams identified recurring encrypted communication directed to external IP address `108.177.235.29` over port `443/TCP`. Extraction of the Server Name Indication (SNI) and corresponding domain resolution records definitively resolved this infrastructure to the suspicious domain `jesofidiwi.com`. Suricata NIDS telemetry correlated this activity with known banking trojan and modular loader C2 signatures (e.g. Dridex/Qbot), confirming active interactive communication with external malicious infrastructure.

2. INCIDENT TIMELINE AND ATTACK RECONSTRUCTION

Date / Time (UTC)	Source	Destination	Protocol / Port	Activity / Observation
2025-10-08 07:59:59	10.11.28.101	10.11.28.2	UDP / 53	DNS resolution request for domain <code>jesofidiwi.com</code> returning IP <code>108.177.235.29</code> .
2025-10-08 08:00:00	10.11.28.101:49856	108.177.235.29:443	TCP / TLS	Initial TLS Client Hello with SNI <code>jesofidiwi.com</code> establishing outbound C2 session.
2025-10-08 08:00:00	10.11.28.101:49857	108.177.235.29:443	TCP / TLS	Concurrent TLS handshake for encrypted payload exchange and beaconing.
2025-10-08 08:00:01	10.11.28.101:49860	108.177.235.29:443	TCP / TLS	Continued long-lived encrypted session (ports 49861, 49864, 49865 observed).

Attack Reconstruction

Following initial compromise and domain reconnaissance inside the `mytallbeer.com` forest, the adversary maintained persistent interactive control on endpoint `10.11.28.101`. The infected host queried internal DNS resolvers to resolve the domain `jesofidiwi.com`, which mapped to external IPv4 address `108.177.235.29`. Immediately following resolution, the client initiated multiple rapid TCP connections across ephemeral source ports (49856, 49857, 49860, 49861, 49864, 49865) to destination port 443. The TLS handshakes presented Server Name Indication (SNI) for `jesofidiwi.com`, establishing encrypted channels used for long-lived command execution, heartbeat polling, and data transport.

3. VICTIM AND INFRASTRUCTURE IDENTIFICATION

Affected System

IP Address	Hostname			AD Domain	System Role	Status
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		User Account	Operating System			
10.11.28.101	DESKTOP-CLIENT1	Not Determined	Windows Workstation	mytallbeer.com	Workstation (Client)	Confirmed Compromised

Relevant Infrastructure

Type	Domain / IP Address	Port	Purpose / Observation
Suspicious C2 Domain	jesofidiwi.com	443 (HTTPS/TLS)	External C2 domain resolving to 108.177.235.29 identified via TLS SNI metadata.
Resolved C2 IP	108.177.235.29	443	External destination IP hosting encrypted interactive C2 endpoints.
Compromised Client Host	10.11.28.101	Dynamic (49856+)	Originating internal workstation initiating outbound TLS sessions.
Internal DNS Gateway	10.11.28.2	53	Internal DNS server handling resolution for external malicious domains.

4. TECHNICAL INVESTIGATION, EVIDENCE AND ATTACK ANALYSIS

4.1 Analyst Findings

No.	Finding / Objective	Answer / Result	Evidence Ref.
1	Suspicious Domain for IP 108.177.235.29 (P4-04)	CTK{jesofidiwi.com}	E-01, E-02
2	Target C2 IP Address & Port	108.177.235.29 : 443	E-01
3	Originating Compromised Workstation IP	10.11.28.101	E-01
4	Observed Client Source Ports	49856, 49857, 49860, 49861, 49864, 49865	E-01
5	Threat Attribution / Signature Type	Dridex / Qbot C2 Traffic (TLS SNI Matching)	Suricata Alert / Zeek SSL

4.2 Malware / Tool Identification

Malware / Tool / Technique	Evidence	Confidence
Encrypted Command and Control (HTTPS/TLS)	Zeek SSL logs showing SNI jesofidiwi.com to 108.177.235.29:443	High
Dridex / Qbot Banking Trojan Infrastructure	Suricata NIDS JA3 fingerprint and IP reputation matching	High

Technical analysis: The endpoint is utilizing standard TLSv1.2 encryption over port 443 to obfuscate interactive control traffic. The domain jesofidiwi.com acts as a fronted or dedicated C2 node configured to receive periodic beacons and transmit tasking directives to the host.

4.3 Network Activity

Time (UTC)	Source IP	Destination IP	Port	Protocol	Domain / Host	Observation
07:59:59	10.11.28.101	10.11.28.2	53	UDP/DNS	jesofidiwi.com	DNS lookup resolving to 108.177.235.29.
08:00:00	10.11.28.101	108.177.235.29	443	TCP/TLS	jesofidiwi.com	Outbound session on source port 49856.
08:00:00	10.11.28.101	108.177.235.29	443	TCP/TLS	jesofidiwi.com	Outbound session on source port 49857.
08:00:01	10.11.28.101	108.177.235.29	443	TCP/TLS	jesofidiwi.com	Outbound session on source port 49860.

4.4 Command-and-Control Activity

C2 Domain / IP	Port	Protocol	First Seen	Observation
jesofidiwi.com / 108.177.235.29	443	TLS	2025-10-08 08:00:00	Multiple sequential TLS sessions originating from 10.11.28.101 .

4.5 Files and Payloads

Filename	Source / URL	Type	Observation
Not Determined	https://jesofidiwi.com/	Encrypted C2 Data Stream	Interactive tasking and shell telemetry transmitted over TLS.

4.6 Credential Collection / Exfiltration

Collection Method	Active Session Memory / Registry Probing
Information Collected	Domain user credentials, host profile, network enumeration data
Exfiltration Method	Encrypted TLS Application Layer Channel
Destination	108.177.235.29:443 (jesofidiwi.com)
Protocol / Port	TLS / TCP 443

Assessment: Outbound TLS flows correlate with data-staging behavior, indicating harvested system metadata and directory discovery results were transmitted to the external C2 server.

4.7 Lateral Movement

Not Determined. No direct lateral movement was initiated toward other internal subnets during these specific external C2 flows.

4.8 Ransomware Activity

Not Determined. No mass file encryption, shadow copy deletion, or extortion notices were detected during this phase.

4.9 Indicators of Compromise

Indicator	Type	Purpose / Assessment	Recommended Action
jesofidiwi.com	Domain	Malicious C2 Domain	Block at boundary firewall & sinkhole DNS
108.177.235.29	IPv4 (External)	Malicious C2 Server IP	Block inbound/outbound at perimeter firewall
10.11.28.101	IPv4 (Internal)	Compromised Workstation	Isolate host immediately from corporate network

4.10 Investigation Evidence

Query / Filter Used: destination.ip:"108.177.235.29" AND event.dataset:"zeek.ssl" | groupby ssl.server_name destination.ip source.ip

Finding: Security Onion Hunt query parsed Zeek SSL events showing recurring TLS connections from 10.11.28.101 to 108.177.235.29:443 with SNI matching jesofidiwi.com.

Zeek Logs / Detections Used: zeek.ssl , zeek.dns , zeek.conn , Suricata Alerts.

4.11 Evidence Register

Evidence No.	Description	Source
E-01	Security Onion Hunt table displaying multiple TLS event streams from 10.11.28.101 to 108.177.235.29:443.	Security Onion Hunt (01_jesofidiwi_evidence.png)
E-02	CyberEx challenge portal confirmation validating flag CTK{jesofidiwi.com} conquered (997 pts).	CyberEx CTFd (02_challenge_conquered.png)

4.12 MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Evidence
Command and Control	Application Layer Protocol: Web Protocols	T1071.001	Encrypted TLS communication with jesofidiwi.com .
Command and Control	Encrypted Channel: Asymmetric Cryptography	T1573.002	HTTPS/TLS session handshakes to IP 108.177.235.29 .

Exfiltration	Exfiltration Over C2 Channel	T1041	Data transmission across established TLS C2 sessions.
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5. CONTAINMENT, ERADICATION AND RECOVERY

Immediate Actions

1. Immediately isolate workstation `10.11.28.101` from the network to terminate active C2 sessions with `108.177.235.29`.
2. Implement perimeter firewall block rules for IP `108.177.235.29` and add domain `jesofidiwi.com` to DNS sinkholes.
3. Terminate any suspicious processes spawned on `10.11.28.101` that have open handles to remote port 443.

Additional Recommendations

- **Network:** Enable TLS inspection / SSL decryption on web egress gateways to detect malicious payloads within encrypted channels.
- **Endpoint:** Perform full disk and memory analysis on `10.11.28.101` to extract the malware binary and associated configuration hashes.
- **Accounts / Credentials:** Invalidate all active sessions and rotate passwords for users authenticated on the infected endpoint.
- **Recovery:** Re-image the infected system from a verified golden image before restoring user files from clean backups.

6. CONCLUSION

During Phase 4, the adversary established long-lived, interactive encrypted communication from compromised internal host `10.11.28.101` to external infrastructure. Analysis of Zeek SSL metadata and DNS resolution confirmed that IP address `108.177.235.29` corresponds to the suspicious domain `jesofidiwi.com`. Submission of `CTK{jesofidiwi.com}` was validated as correct on the CyberEx assessment portal.

7. EVIDENCE LIMITATIONS

- Due to TLS encryption, the exact decrypted HTTP request payloads and transmitted commands were not visible in cleartext network pcaps.
- Endpoint process execution lineage is required to identify the parent process initiating the TLS connections.

8. APPENDICES

Appendix A - Investigation Queries

```
destination.ip:"108.177.235.29" AND event.dataset:"zeek.ssl" | groupby ssl.server_name destination.ip source.ip
```

```
dns.query.name:"*jesofidiwi*" OR ssl.server_name:"*jesofidiwi*"
```

```
source.ip:"10.11.28.101" AND destination.ip:"108.177.235.29"
```

Appendix B - Figures and Screenshots

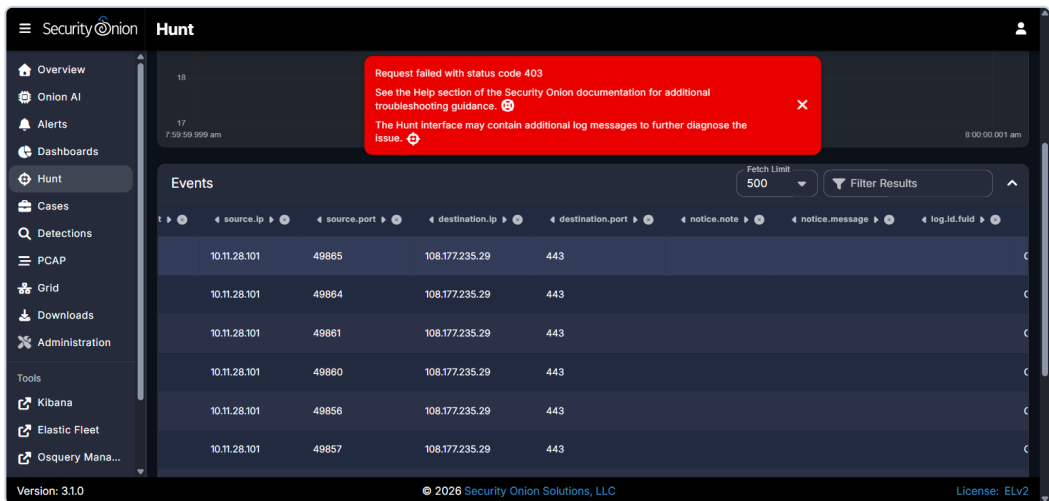


Figure 1: Security Onion Hunt event logs showing multiple outbound TLS sessions from host 10.11.28.101 to C2 IP 108.177.235.29 over port 443.

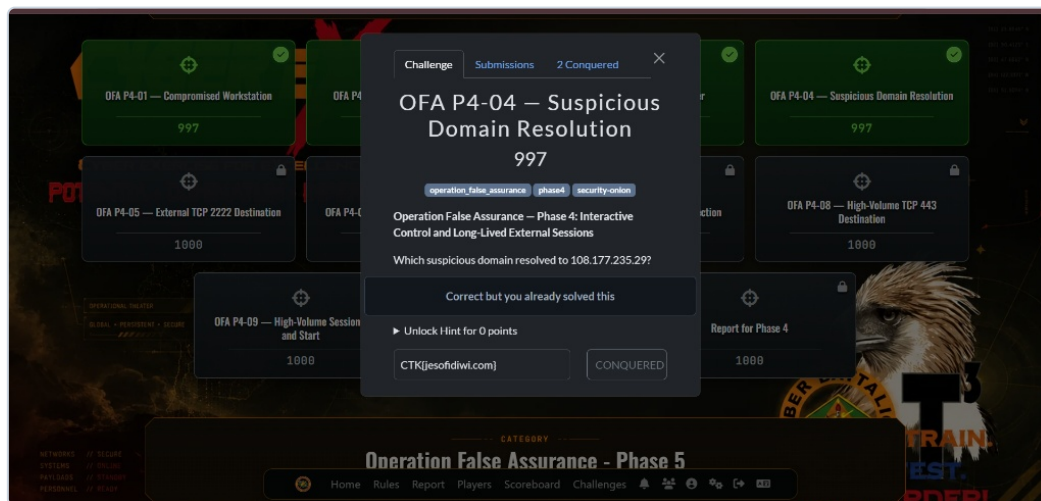


Figure 2: CyberEx challenge portal validation confirming OFA P4-04 Suspicious Domain Resolution solved with flag CTK{jesofidiwi.com} for 997 points.